Exercise on Group Selection

- There are 7 boys and 5 girls in a class. We want to form a 4-person group from these 12 people, so that there is at least 1 girl in the group. In how many ways can the 4-person group be chosen?
- Solution:
  - Select without any constraint C(12,4)=495.
  - Select only boys C(7,4)=35.
  - Select at least one girl: C(12,4)- C(7,4)=460.
- Alternative Solution: The group may contain:
  - 4 girls in C(5,4)=5 ways, or
  - 3 girls and 1 boy in C(5,3)xC(7,1)=10x7=70 ways, or
  - 2 girls and 2 boys in C(5,2)xC(7,2)=10x210=210 ways, or
  - 1 girl and 3 boys in C(5,1)xC(7,3) = 5x35 = 175 ways.
  - The sum of all ways is 460.